

Venkata Ramakrishna Kamepalli

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PROFESSIONAL SUMMARY

Python-proficient Software Engineering student (M.Tech, VIT Chennai | CGPA: 8.27/10) with hands-on experience designing, coding, and enhancing software components across full-stack and cloud-based systems using modern frameworks and agile methodologies. Demonstrated ability to write clean, maintainable, and efficient Python code following object-oriented programming principles and software development best practices. Experienced in version control (Git), debugging, root cause analysis, unit testing (Pytest), and cross-functional team collaboration. MongoDB Certified Associate Developer. Certified in Anthropic Claude API, MCP, and Agent Skills. Strong communicator with a continuous learning mindset

EDUCATION

Vellore Institute of Technology (VIT) <i>Integrated M.Tech in Software Engineering CGPA: 8.27/10 15 Years Full-Time Education</i>	Chennai, India 2022 – 2027
Saraswathi Junior College <i>Intermediate (Class XII) 97%</i>	Ongole, India 2020 – 2022
Apex High School <i>Secondary School (Class X) 98%</i>	Ongole, India 2019 – 2020

TECHNICAL SKILLS

Programming Languages: Python (Primary), Java, JavaScript (ES6+), TypeScript, SQL, C
Python Ecosystem: Pytest, OOP, Multi-threading, Debugging, Root Cause Analysis, Scripting, Automation
Software Development: Object-Oriented Programming, Clean Code, Agile/Scrum, Sprint Planning, Code Reviews
Version Control & DevOps: Git (GitHub), Docker, CI/CD fundamentals
Cloud & Backend: AWS (Lambda, API Gateway, EC2, S3, SNS, CloudWatch, Cognito, DynamoDB), Node.js, Express.js, REST APIs
AI/ML & Gen-AI: Generative AI, RAG, LLM Fundamentals, CNN, Anthropic Claude API, MCP, Agent Skills
Databases: MongoDB (Certified Associate Developer), PostgreSQL, MySQL, DynamoDB
Frameworks: React.js, Next.js, Spring Framework, Hibernate

PROJECTS

- StudyMate – AI-Powered Notes Application** | *Python, AWS Lambda, S3, API Gateway, Cognito, CloudWatch, GenAI2023 – 2024*
- Designed and coded a production-grade serverless AI application in **Python** on AWS, delivering 5 LLM-integrated features — MCQ generation, Summarization, Flashcards, Mindmaps, and Detailed Notes — from uploaded study material, following scalable software architecture principles.
 - Developed clean, maintainable Python Lambda functions forming an event-driven pipeline (Lambda → API Gateway → S3 → EC2) with prompt engineering for consistent, structured AI output — applying OOP and modular design best practices.
 - Optimised code-level performance and cloud resource tuning to **reduce API latency by 20%**; implemented AWS Cognito authentication and CloudWatch monitoring, demonstrating debugging and troubleshooting skills to resolve runtime issues.
 - Executed **end-to-end unit testing** across all Python AI feature modules before deployment — ensuring software quality, reliability, and production readiness at scale.
- IoT-Based Scalp Disease Detection System** | *Python, Pytest, CNN, Embedded C++, ESP32-CAM, Linux2023 – 2024*
- Authored **Python automation scripts** integrating a CNN deep learning model into an embedded hardware pipeline, enabling intelligent real-time disease classification — end-to-end custom software development across system layers.
 - Designed and executed **Pytest-based automated test suites** for sensor data validation and pipeline integrity — achieving robust verification coverage across all components, applying structured debugging and root cause analysis.
 - Implemented **multi-threaded concurrent processing** in Python on Linux/Ubuntu; resolved 5+ hardware-software integration bugs through systematic root cause analysis, **improving system reliability by 40%**.
 - Documented software design decisions and integration processes to facilitate team understanding and future maintenance — directly supporting cross-functional collaboration.

VIT Connect – Student Community Platform | *React.js, Node.js, Express.js, MongoDB, REST APIs, Git, Agile*2023

- Designed and built a full-stack MERN application with component-based architecture, RESTful backend APIs, and MongoDB NoSQL storage — delivering clean, maintainable code aligned with industry best practices.
- Optimised MongoDB query performance by **20%** through indexing and aggregation pipeline tuning; implemented JWT-based authentication and scalable REST API design.
- Led a **4-member Agile team** through sprint planning, daily stand-ups, and Git-based peer code reviews — gathering requirements, providing technical insights, and delivering the platform within timeline.
- Managed the full codebase using **Git version control**, enforcing branching strategies and review workflows to ensure code quality and team-level maintainability.

HR Workflow Designer | *TypeScript, React.js, Node.js, Express.js, REST APIs, Git*2024 – Present

- Architected a type-safe full-stack workflow automation tool with drag-and-drop React.js frontend and Node.js/Express.js backend — applying modular architecture, clean code principles, and comprehensive documentation.
- Applied agile development practices, Git-based version control, and iterative sprint-based delivery to build scalable, high-performing software components tailored to specific business workflow needs.

CERTIFICATIONS

MongoDB Associate Developer <i>MongoDB</i>	May 2026 <i>Verify: credly.com/users/venkata-ramakrishna.fe20e510</i>
Claude with the Anthropic API <i>Anthropic</i>	May 2026 <i>Verify: verify.skilljar.com/c/y74wjp348erp</i>
Introduction to Model Context Protocol (MCP) <i>Anthropic</i>	May 2026 <i>Verify: verify.skilljar.com/c/t832agpz7q82</i>
Introduction to Agent Skills <i>Anthropic</i>	May 2026 <i>Verify: verify.skilljar.com/c/gj7g8nr79xug</i>
Python Programming <i>Coursera</i>	2024
Agile Scrum in Practice <i>Infosys Springboard</i>	2024
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate <i>Oracle University</i>	2025

ADDITIONAL INFORMATION

Languages: English (Fluent), Telugu (Native), Hindi (Conversational)
Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Software Engineering, Cloud Computing
Soft Skills: Problem-Solving, Team Collaboration, Cross-functional Communication, Continuous Learning, Documentation
Academic Excellence: 98% in Class X, 97% in Class XII
Availability: Full-time, Work from Office, Pune / Bangalore / Hyderabad / Chennai